

Optimizing Face Verification: Hard Pair Mining and ArcFace Margin Loss

SCENARIO

Application Context: Student Exam Authentication System. The system verifies that the student sitting the exam is the one who registered.

The core research challenge is preventing extreme lookalikes (Hard Negatives) from bypassing the system while ensuring legitimate students with different poses or lighting (Hard Positives) are not wrongly blocked.

OBJECTIVE

Build a face verification system using EfficientNetV2 (S, M, L) via a Siamese Network. The project extends the baseline by replacing standard contrastive loss with ArcFace (Additive Angular Margin Loss) and implementing dynamic Hard Pair Mining, evaluated strictly using the LFW Benchmark.

PIPELINE

1. Data Acquisition: Utilize the standard LFW evaluation protocol, containing 13,233 face images and 6,000 verification pairs.
2. Pre-processing: Apply MTCNN face detection & alignment. Normalize to EfficientNetV2 input sizes (300x300, 384x384, 480x480). Implement a custom data loader for dynamic Hard Pair Mining.
3. Feature Extraction: Use EfficientNetV2 variants fine-tuned from ImageNet pretrained weights to extract 1280-dim embeddings.
4. Matching & Training: Process pairwise inputs via shared weights. Conduct 6 training runs: 3 baseline models using contrastive loss and 3 advanced models using ArcFace + Hard Mining.
5. Decision: Calculate distance using cosine similarity for the similarity decision.

EXPECTED OUTCOMES

- Achieve a target EER < 5% on the LFW benchmark.
- Ensure EfficientNetV2-S delivers fast, sub-second verification for real-time authentication.
- Produce a side-by-side FAR, FRR, EER and latency table.